

# AUDITING GENERATIVE AUDIO CALLS FOR KNOWN-TASK AUDIO-LLM EVALUATION

*Mengzhe Geng*

National Research Council Canada  
Mengzhe.Geng@nrc-cnrc.gc.ca

## ABSTRACT

Speech and audio LLMs are often evaluated by asking whether a waveform prompt beats an automatic speech recognition (ASR) transcript. For known closed-set tasks, that comparison conflates two factors: access to acoustic evidence and the need to call a generative audio model. We evaluate this distinction as a controlled call-decision problem. For each example, a policy chooses among keeping a transcript label, using encoder evidence from Contrastive Language–Audio Pretraining (CLAP), Audio Spectrogram Transformer (AST), or WavLM, and calling Qwen2-Audio, Qwen2.5-Omni, or MOSS-Audio; the decisive ablation removes all generative actions while keeping the selector and development protocol fixed. On VocalSound, transcripts reach 0.296 accuracy, so waveform information is needed. Yet supervised CLAP and WavLM controls reach 0.850 and 0.854 with no generative audio calls. A selector with generative actions reaches 0.925 accuracy using 12.5% calls, compared with 0.921 for the matched no-call selector (paired difference 0.004; 95% CI  $[-0.025, 0.033]$ ). Agreement and stacking features improve weaker selectors but do not beat the strongest no-call control. For known-task endpoint claims, the relevant quantity is the marginal value of the generative call after transcript and encoder evidence have already been used.

**Index Terms**— audio-language models, generative-call audit, audio-LLM evaluation, selective prediction, cascaded inference

## 1. INTRODUCTION

Recent speech and audio LLMs take waveforms as input and return text decisions, descriptions, or dialogue responses [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]. They are useful when automatic speech recognition (ASR) text discards information carried by the signal, such as vocalizations, emotion, speaker state, or overlapping speech [12, 13]. A common evaluation compares an audio-LLM prediction with a transcript-only prediction. That test shows whether the waveform contains useful evidence, while leaving the interface question open: did the system need a generative audio model to use that evidence?

In a deployed closed-set task, a system chooses among

keeping a transcript-side label, scoring the waveform with a local encoder, and sending the waveform to a generative audio model. The last option sometimes improves decisions, but it also adds cost and, in hosted settings, increases exposure of speech data. We ask whether generative audio calls still change the answer enough to justify their use after transcript and encoder evidence are available to the same development-selected policy.

We answer this question with locked row splits and matched controls. The evaluation covers Qwen2-Audio, Qwen2.5-Omni, MOSS-Audio, Contrastive Language–Audio Pretraining (CLAP), Audio Spectrogram Transformer (AST), and WavLM; compares transcript-first, encoder-first, full-selector, and no-call policies; and reports paired intervals, Holm-adjusted tests, and measured sequential costs. The conclusion is intentionally limited to known closed-set label decisions. We do not evaluate open dialogue quality, instruction-following ability, or general audio reasoning.

## 2. RELATED WORK

SUPERB evaluates reusable speech representations, and Dynamic-SUPERB extends this evaluation style to collaborative instruction-following speech tasks [14, 15]. Recent audio-language benchmarks add generative comprehension, voice-assistant behavior, hallucination, egocentric multi-talker speech, and speaker understanding [16, 17, 18, 19, 20, 21]. These benchmarks primarily measure model ability. Our evaluation instead fixes a known task and asks whether a generative audio call still changes the decision after transcript and encoder controls are included.

The comparison also draws on calibration, misclassification detection, selective prediction, ASR confidence, and LLM cascades [22, 23, 24, 25, 26]. Transcript cascades usually decide when to leave text. Here, audio encoders supply first-stage evidence. We include CLAP, AST, and WavLM controls [27, 28, 29] and compare routed generative calls with selectors that see the same encoder evidence but never invoke a generative audio model.

### 3. GENERATIVE-CALL DECISION TEST

For utterance  $x_i$ , ASR produces transcript  $y_i$ , and a text LLM maps that transcript to label  $\hat{c}_i^{\text{text}}$ . A generative audio model predicts  $\hat{c}_i^{\text{gen}}$  directly from the waveform. Given a target call budget  $b$ , a routing policy selects examples  $\mathcal{R}_b$  for the generative model and returns

$$\hat{c}_i = \begin{cases} \hat{c}_i^{\text{gen}}, & i \in \mathcal{R}_b, \\ \hat{c}_i^{\text{text}}, & i \notin \mathcal{R}_b, \end{cases} \quad (1)$$

where  $\mathcal{R}_b = \{i : s_i \geq \tau_b\}$  and  $\tau_b$  is chosen on development rows. The transcript routing score is

$$s_i = \rho_i^{\text{text}} + \alpha u_i + \beta q_i + \gamma v_i, \quad (2)$$

where  $\mathcal{C}$  is the task label set,  $\rho_i^{\text{text}}$  is the development-estimated risk of the transcript label,  $u_i$  is normalized Whisper uncertainty,  $q_i$  is normalized text-LLM label-likelihood uncertainty, and  $v_i$  is one when  $\hat{c}_i^{\text{text}} \notin \mathcal{C}$ . We choose the weights and threshold on development rows only. We also test an encoder-first route: CLAP scores every waveform and sends low-margin examples to a generative audio model. Supervised CLAP, WavLM, and AST probes use labeled development rows; we report them as deployment controls.

The full selector is a stronger test of call value. It fits an L2-regularized logistic correctness model over row-action pairs, using action confidence, transcript uncertainty and length, encoder flags, generative-call flags, action cost, and action identity. CLAP and WavLM development predictions are out-of-fold. We choose  $\lambda$  on development accuracy, breaking ties toward fewer generative calls. The matched no-call selector uses the same training and selection protocol after removing Qwen2 and Omni actions. Qwen2 and Omni do not expose comparable token-level confidence in the cached outputs, so their confidence feature is constant; the full-vs-no-call comparison tests whether the available decision traces justify a generative action. Agreement features and a ridge classifier over one-hot action predictions serve as diagnostics.

Before evaluating the holdout rows, we fix the parser, label normalization, features, weights, thresholds, and  $\lambda$  on development rows. Prompts, parser code, model identifiers, frozen outputs, and selector files are kept for reproduction, and all settings are applied unchanged to the holdout. Paired intervals use 10k row bootstraps, McNemar  $p$  values are exact, point intervals use Wilson intervals, and the eight headline paired tests use Holm correction. A 10-seed diagnostic varies CLAP/WavLM folds and selector fitting while keeping generative labels fixed.

Costs are reported separately from accuracy. For a transcript-first route, the measured sequential cost is

$$C_i^{\text{route}} = C_i^{\text{asr}} + C_i^{\text{text}} + h_i C_i^{\text{conf}} + r_i C_i^{\text{gen}}, \quad (3)$$

where  $h_i$  marks optional transcript-confidence scoring and  $r_i$  marks a generative audio call. For a CLAP-first policy, the

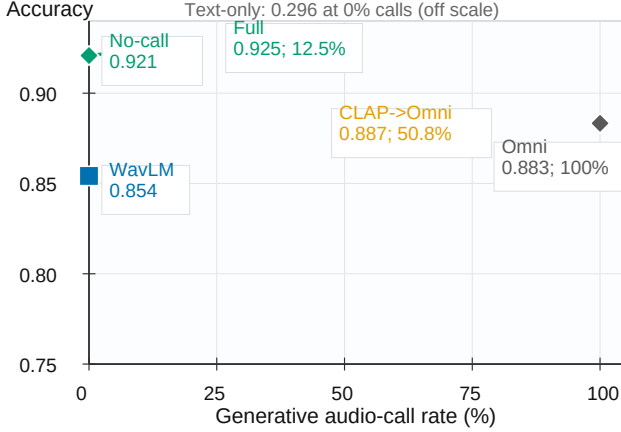
analogous cost is  $C_i^{\text{clap}} + r_i C_i^{\text{gen}}$ : every waveform reaches CLAP, but only routed examples reach the generative audio model. This accounting measures call reduction. Privacy and serving latency require separate measurement, because hosted systems expose transcripts, embeddings, prompts, or routed waveforms according to their interface, and first-stage overhead affects unbatched sequential execution.

### 4. EVALUATION

The main benchmark is Dynamic-SUPERB VocalSound, a six-way human-vocalization task with chance accuracy 0.167 and labels for laughter, sigh, cough, sneeze, sniff, and throat clearing [30]. Rows 0–479 form the development split, and rows 480–719 form the same-dataset holdout. A filename-derived check finds no overlap in file stems, inferred clip groups, inferred speaker IDs, or inferred session IDs; we treat the split as a locked row holdout. We also evaluate ESC-50 Animals ( $n = 200$ ) [31], a five-emotion AudioBench paralinguistic question-answering (PQA) subset drawn from the English portion of Emotional Speech Dataset (ESD), with development  $n = 90$  and holdout  $n = 120$  [17, 32], and Dynamic-SUPERB SpeechTextMatching LibriSpeech-TestClean (STM;  $n = 100$ ) over LibriSpeech [15, 33]. A metric checker recomputes every headline number from stored summaries.

VocalSound first confirms that transcripts are insufficient: the transcript-only system reaches 0.296 accuracy. Audio models are much stronger: Qwen2.5-Omni reaches 0.883, Qwen2-Audio 0.838, MOSS-Audio 0.808, zero-shot CLAP 0.762, and AudioSet AST 0.404. Development-supervised encoder controls narrow the gap further. CLAP class-bias tuning reaches 0.792, a CLAP-score probe reaches 0.796 accuracy and 0.784 macro F1, a CLAP-embedding probe reaches 0.850/0.849, AST embeddings reach 0.817/0.813, and WavLM-base+ reaches 0.854/0.853. A CLAP-WavLM correctness oracle reaches 0.938, and a joint CLAP-WavLM ridge probe reaches 0.896/0.892. These controls move the comparison from transcript-versus-audio accuracy to the marginal value of a generative audio call. The call-rate frontier is shown in Fig. 1; representative policy costs and paired accuracy contrasts are listed in Table 1.

The paired comparisons put the main claims on matched rows. CLAP-first Qwen2-Audio reaches 0.850 accuracy with 50.8% generative calls, matching the supervised CLAP-embedding probe but with lower macro F1 (0.841 versus 0.849) and higher call cost. CLAP-first Qwen2 and CLAP-first MOSS are slightly below WavLM-emb. The best routed policy sends low-margin CLAP examples to Omni and reaches 0.887 with 50.8% generative calls. Its paired advantage over WavLM-emb is not detected ( $\Delta = 0.033$ , 95% CI  $[-0.017, 0.083]$ , McNemar  $p = 0.268$ ). The interval leaves open both no gain and a useful gain, so this comparison remains inconclusive. Prompt sensitivity leaves 50% CLAP-



**Fig. 1.** VocalSound row-holdout accuracy versus generative-call rate for representative policies. The main axis is cropped to the competitive region; the transcript-only baseline is annotated above the panel. The x-axis counts only generative audio-model calls; cheap controls still process transcripts or local audio features. The full selector is highest, but its paired advantage over the no-call selector includes zero.

first Qwen2 accuracy between 0.792 and 0.850, making label wording and encoder access part of the result.

Cost accounting changes the interpretation of selector results. After excluding irrelevant transcript scoring, the no-call selector uses 11.3 s of cached sequential component time; the earlier accounting assigns 45.8 s. The full selector uses 16.9 s after selected generative calls are added, down from 51.4 s under the earlier accounting. The direct full-vs-no-call comparison is 0.925 versus 0.921,  $\Delta = 0.004$ , CI  $[-0.025, 0.033]$ . Agreement-aware and stacked selectors give positive within-family deltas, but both remain below the strongest no-call policy in cross-family comparisons ( $\Delta = -0.021$  and  $-0.008$ ). Across eight headline paired tests (five in Table 1 and three on ESD), only raw full-vs-WavLM survives Holm correction ( $p_{\text{adj}} = 0.027$ ); agreement and stacked within-family p-values adjust to 0.154 and 0.091. Across 10 fold seeds, agreement-full accuracy is  $0.897 \pm 0.005$  and full-minus-no-call is  $0.042 \pm 0.007$ .

The external checks point in the same direction. ESC-50 Animals confirms the value of acoustic evidence over text alone (CLAP 0.895, Qwen2-Audio 0.785, transcript-only 0.140). On fresh ESD, Qwen2-Audio reaches 0.425 accuracy and 0.341 macro F1, WavLM reaches 0.475/0.444, and the raw full-vs-no-call comparison gives 0.433 versus 0.425 ( $\Delta = 0.008$ , CI  $[-0.067, 0.083]$ ). The agreement diagnostic is negative on ESD (0.442 versus 0.475,  $\Delta = -0.033$ , CI  $[-0.075, 0.008]$ ). Stacked fusion is positive (0.550 versus 0.475,  $\Delta = 0.075$ , CI  $[0.017, 0.133]$ ) but not Holm-significant. STM reverses the ordering, with transcript accuracy 0.960 versus Qwen2-Audio 0.850, so call decisions are task-specific.

**Table 1.** VocalSound holdout summary ( $n = 240$ ). Rate is the generative-call fraction; Total and Gen. are cached sequential seconds. The paired block uses each row’s stated no-call baseline.

| Representative policies |              |       |       |      |
|-------------------------|--------------|-------|-------|------|
| Policy                  | Acc.         | Rate  | Total | Gen. |
| Text                    | 0.296        | 0.000 | 34.4  | 0.0  |
| WavLM-emb               | 0.854        | 0.000 | 2.5   | 0.0  |
| Qwen2                   | 0.838        | 1.000 | 31.5  | 31.5 |
| CLAP→Omni               | 0.887        | 0.508 | 30.5  | 21.7 |
| No-call sel             | 0.921        | 0.000 | 11.3  | 0.0  |
| Full sel                | <b>0.925</b> | 0.125 | 16.9  | 5.6  |

| Paired accuracy deltas |          |                   |
|------------------------|----------|-------------------|
| Contrast               | $\Delta$ | 95% CI            |
| CLAP→Omni-WavLM        | 0.033    | $[-0.017, 0.083]$ |
| Full-WavLM             | 0.071    | $[0.029, 0.117]$  |
| Raw full-no-call       | 0.004    | $[-0.025, 0.033]$ |
| Agree full-no-call     | 0.042    | $[0.008, 0.079]$  |
| Stack full-no-call     | 0.042    | $[0.013, 0.075]$  |

## 5. DISCUSSION AND LIMITATIONS

The evaluation inherits the biases of ASR systems, audio encoders, and LLMs on accents, dialects, disordered speech, code-switching, and low-resource languages. Reducing generative calls also leaves some audio exposure: CLAP-first and full selectors still process transcripts and encoder features, and hosted systems expose routed waveforms or intermediate summaries according to their interface. The VocalSound split is locked for this study, but repeated use will turn it into a fixed test set. Fresh ESD is row-disjoint but small; a larger speaker-disjoint rerun is the most important next experiment. The result is task-scoped as well. Supervised CLAP/WavLM embeddings match or exceed routed generative calls here; richer reasoning tasks, unknown task mixtures, few-shot adaptation, calibrated generative log-probabilities, and throughput-aware serving define separate tradeoffs. Deployment use requires consent, privacy review, fairness analysis, and strict data minimization.

## 6. CONCLUSION

An audio-vs-transcript gain is insufficient evidence that a generative audio call was necessary. In these locked experiments, transcripts fail on non-speech vocalizations, supervised encoders are strong when development labels exist, and the best no-call selector nearly ties a full selector with access to generative audio models. Agreement and stacking gains depend on selector design and fall short of a stable endpoint-call result. Future reports need an explicit call boundary: what is decided from transcripts, what is decided from local encoders, and what changes only after the waveform reaches the generative model.

## 7. REFERENCES

- [1] Dong Zhang, Shimin Li, Xin Zhang, Jun Zhan, Pengyu Wang, Yaqian Zhou, and Xipeng Qiu, “Speechgpt: Empowering large language models with intrinsic cross-modal conversational abilities,” 2023.
- [2] Paul K. Rubenstein, Chulayuth Asawaroengchai, Duc Dung Nguyen, Ankur Bapna, Zalán Borsos, Félix de Chaumont Quitry, Peter Chen, Dalia El Badawy, et al., “Audiopalm: A large language model that can speak and listen,” 2023.
- [3] Changli Tang, Wenyi Yu, Guangzhi Sun, Xianzhao Chen, Tian Tan, Wei Li, Lu Lu, Zejun Ma, et al., “Salmonn: Towards generic hearing abilities for large language models,” 2023.
- [4] Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, Yichong Leng, Yuanjun Lv, et al., “Qwen2-audio technical report,” 2024.
- [5] Zhifei Xie and Changqiao Wu, “Mini-omni: Language models can hear, talk while thinking in streaming,” 2024.
- [6] Qingkai Fang, Shoutao Guo, Yan Zhou, Zhengrui Ma, Shaolei Zhang, and Yang Feng, “Llama-omni: Seamless speech interaction with large language models,” 2024.
- [7] Aohan Zeng, Zhengxiao Du, Mingdao Liu, Kedong Wang, Shengmin Jiang, Lei Zhao, Yuxiao Dong, and Jie Tang, “Glm-4-voice: Towards intelligent and human-like end-to-end spoken chatbot,” 2024.
- [8] Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, Jialin Wang, et al., “Qwen2.5-omni technical report,” 2025.
- [9] Chen Yang, Chufan Yu, Hanfu Chen, Jie Zhu, Jingqi Chen, Ke Chen, Wenxuan Wang, Yang Wang, et al., “Moss-audio technical report,” 2026.
- [10] Boyong Wu, Chao Yan, Chen Hu, Cheng Yi, Chengli Feng, Fei Tian, Feiyu Shen, Gang Yu, et al., “Step-audio 2 technical report,” 2025.
- [11] Sreyan Ghosh, Arushi Goel, Kaousheik Jayakumar, Lasha Koroshinadze, Nishit Anand, Zhifeng Kong, Siddharth Gururani, Sang-gil Lee, et al., “Audio flamingo next: Next-generation open audio-language models for speech, sound, and music,” 2026.
- [12] Wenqian Cui, Dianshi Yu, Xiaoqi Jiao, Ziqiao Meng, Guangyan Zhang, Qichao Wang, Yiwen Guo, and Irwin King, “Recent advances in speech language models: A survey,” 2024.
- [13] Jing Peng, Yucheng Wang, Bohan Li, Yiwei Guo, Han-kun Wang, Yangui Fang, Yu Xi, Haoyu Li, et al., “A survey on speech large language models for understanding,” 2024.
- [14] Shu-wen Yang, Po-Han Chi, Yung-Sung Chuang, Cheng-I Jeff Lai, Kushal Lakhota, Yist Y. Lin, Andy T. Liu, Jiatong Shi, et al., “Superb: Speech processing universal performance benchmark,” 2021.
- [15] Chien-yu Huang, Ke-Han Lu, Shih-Heng Wang, Chi-Yuan Hsiao, Chun-Yi Kuan, Haibin Wu, Siddhant Arora, Kai-Wei Chang, et al., “Dynamic-superb: Towards a dynamic, collaborative, and comprehensive instruction-tuning benchmark for speech,” 2023.
- [16] Qian Yang, Jin Xu, Wenrui Liu, Yunfei Chu, Ziyue Jiang, Xiaohuan Zhou, Yichong Leng, Yuanjun Lv, et al., “Air-bench: Benchmarking large audio-language models via generative comprehension,” 2024.
- [17] Bin Wang, Xunlong Zou, Geyu Lin, Shuo Sun, Zhuohan Liu, Wenyu Zhang, Zhengyuan Liu, AiTi Aw, et al., “Audiobench: A universal benchmark for audio large language models,” 2024.
- [18] Yiming Chen, Xianghu Yue, Chen Zhang, Xiaoxue Gao, Robby T. Tan, and Haizhou Li, “Voicebench: Benchmarking llm-based voice assistants,” *Transactions of the Association for Computational Linguistics*, vol. 14, pp. 378–398, 2026.
- [19] Feiyu Zhao, Yiming Chen, Wenhuan Lu, Daipeng Zhang, Xianghu Yue, and Jianguo Wei, “Halluaudio: A comprehensive benchmark for hallucination detection in large audio-language models,” in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, San Diego, California, United States, 2026, pp. 38797–38816, Association for Computational Linguistics.
- [20] Dehui Gao, Zhixian Zhao, Zhennan Lin, Yujie Liao, Yuhang Dai, Yike Zhu, Longshuai Xiao, Hui Bu, et al., “The slt 2026 smartglasses challenge: Benchmarking egocentric multi-talker speech recognition and understanding with audio-language models,” 2026.
- [21] Junkai Wu, Xulin Fan, Bo-Ru Lu, Xilin Jiang, Nima Mesgarani, Mark Hasegawa-Johnson, and Mari Ostendorf, “Just asr + llm? a study on speech large language models’ ability to identify and understand speaker in spoken dialogue,” 2024.
- [22] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger, “On calibration of modern neural networks,” in *Proceedings of the 34th International Conference on Machine Learning*, 2017, pp. 1321–1330.

- [23] Dan Hendrycks and Kevin Gimpel, “A baseline for detecting misclassified and out-of-distribution examples in neural networks,” in *International Conference on Learning Representations*, 2017.
- [24] Yonatan Geifman and Ran El-Yaniv, “Selective classification for deep neural networks,” in *Advances in Neural Information Processing Systems*, 2017, vol. 30.
- [25] Dan Oneata, Alexandru Caranica, Adriana Stan, and Horia Cucu, “An evaluation of word-level confidence estimation for end-to-end automatic speech recognition,” in *IEEE Spoken Language Technology Workshop (SLT)*, 2021.
- [26] Lingjiao Chen, Matei Zaharia, and James Zou, “Frugal-gpt: How to use large language models while reducing cost and improving performance,” *Transactions on Machine Learning Research*, 2024.
- [27] Yusong Wu, Ke Chen, Tianyu Zhang, Yuchen Hui, Marianna Nezhurina, Taylor Berg-Kirkpatrick, and Shlomo Dubnov, “Large-scale contrastive language-audio pre-training with feature fusion and keyword-to-caption augmentation,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- [28] Yuan Gong, Yu-An Chung, and James Glass, “AST: Audio spectrogram transformer,” in *Interspeech*, 2021, pp. 571–575.
- [29] Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, Xiangzhan Yu, and Furu Wei, “WavLM: Large-scale self-supervised pre-training for full stack speech processing,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505–1518, 2022.
- [30] Yuan Gong, Jin Yu, and James Glass, “Vocalsound: A dataset for improving human vocal sounds recognition,” 2022.
- [31] Karol J. Piczak, “ESC: Dataset for environmental sound classification,” in *Proceedings of the 23rd ACM International Conference on Multimedia*, 2015, pp. 1015–1018.
- [32] Kun Zhou, Berrak Sisman, Rui Liu, and Haizhou Li, “Emotional voice conversion: Theory, databases and ESD,” *Speech Communication*, vol. 137, pp. 1–18, 2022.
- [33] Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur, “LibriSpeech: An ASR corpus based on public domain audio books,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2015, pp. 5206–5210.